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A general framework for using two-stage meta-analysis with individual participant data to predict individualized treatment effects

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Abstract

Background

A key application of an individual participant data meta-analysis (IPD-MA) of randomized controlled trials is to predict individualized treatment effects (ITEs). A two-stage approach is often employed, for example when data from trials reside in different servers; however, analyses are typically limited to regression-based models. We hereby present a general framework for two-stage IPD-MA for predicting ITE for a continuous outcome, applicable to any type of statistical or machine learning model. We also present a simulation study to assess the performance of the methods described here.

Methods

We outline different types of models that can be used for trial-specific analyses at the first stage, including ordinary least squares (OLS) and penalized regressions, machine learning (ML) models, and Bayesian models. Subsequently, we describe alternative methods to synthesize results at the second stage and predict treatment effects on new individuals. The second-stage methods were chosen based on the first-stage model used, and include: (a) multivariate meta-analyses of regression coefficients, applicable to OLS; (b) weighted average of predicted ITEs, applicable to all models, with weights obtained via either bootstrapping or via auxiliary OLS models; (c) multivariate Bayesian meta-analysis, applicable to Bayesian regressions; (d) weighted mixing of posterior distributions, applicable to all Bayesian models. A simulation study assessed the performance of all feasible combinations of first- and second-stage approaches under various scenarios in terms of median absolute error of predicted ITEs.

Results

The performance of methods in simulations depended on the sample size, the heterogeneity of treatment effects, and the assumed form of effect modification. In scenarios with complex

treatment-covariate interactions and large samples, ML models substantially outperformed regression methods. Conversely, in scenarios with linear effect modification, regression models performed better. Among second-stage methods for pooling ML models, we found that weighting based on an auxiliary OLS model was better than bootstrapping. We found no important differences between frequentist and Bayesian regression-based methods. Shrinkage models provided only modest improvements over unpenalized regression.

Conclusions

Our framework for predicting patient-level treatment effects using two-stage IPD-MA can accommodate any type of statistical or machine learning model. The optimal method to use depends on the data generating mechanism, which in practice is unknown, and should be decided following internal validation.

Keywords: Individualized treatment effect, Individual participant data meta-analysis, personalized medicine, clinical prediction modelling.

1 Background

An individual participant data meta-analysis (IPD-MA) of randomized controlled trials (RCTs) works with raw participant-level data from multiple trials, and aims to make inference regarding the effects of an experimental treatment over a control [1,2]. Advantages of IPD-MA over aggregated data meta-analysis are numerous and well-established [3]. IPD-MAs may improve the quality of the evidence synthesis compared to aggregated data meta-analyses, as access to IPD allows the inclusion of more detailed information [3–5] and can help avoid reporting bias [2]. The use of participant-level information enables also the harmonization of outcomes and covariates definitions across trials [4,6] and allows analyses to be standardized and potentially improved compared to those reported in the original publications, e.g., via adjusting for prognostic factors [1] or appropriate handling of missing data [7,8]. These advantages have contributed to widespread use of IPD-MAs, despite the considerable effort required to obtain the raw data and the additional statistical expertise required to perform the analyses [1].

A key application of IPD-MA is the prediction of individual-level relative effects of an intervention versus a reference treatment, for a given health outcome [9]. This is often termed the individualized treatment effect (ITE), defined as the difference between counterfactual outcomes, in treatment and control [10]; in simple words, ITE represents the benefit that an individual is expected to obtain from an experimental treatment compared with the control intervention. Analyses of ITE shifts the focus from population-level treatment effects, which are the usual aim of evidence synthesis, to treatment effects at the participant level. Results from such analyses can be used to personalize treatment decisions (related to “stratified”, or “precision”, medicine) [9,11]. IPD-MA is ideal for this type of analyses, as RCTs are generally powered to detect average, rather than individual, level effects; consequently, pooling data from multiple studies is often necessary to achieve the required statistical power.

Many IPD-MA applications use one-stage approaches: in this type of analysis, IPD from all RCTs are combined into a single dataset and, subsequently, jointly analysed [12]. However, it might happen that a one-stage approach may be difficult or even infeasible, e.g. if data are subjected to data sharing constraints. In such cases, IPD from different trials may be stored in independent servers, cannot be combined across trials, and therefore must be analysed in a decentralized way [13,14]. This setting naturally leads to a two-stage meta-analysis, in which each study is first analysed separately at the first stage, and results are pooled at a second stage [12]. Even when one-stage approaches are feasible, a two-stage approach may still offer some additional advantages over one-stage methods, including lower computational burden [15] and improved control over the analysis pipeline [2]. For such reasons, interest in two-stage IPD models is increasing.

In this work, we present a general framework for two-stage IPD-MA to predict ITE, with a focus on a continuous health outcome. We consider a range of statistical models, both frequentist and Bayesian, as well as more complex approaches from machine learning (ML). These models are applied independently within each trial at the first stage. We then describe various methods that can be used for pooling results at the second stage, depending on the first-stage model chosen. Finally, we present a simulation study aiming to assess the advantages and disadvantages of each combination of model (at first stage) and pooling method (at second stage), across a range of scenarios.

2 Methods

2.1 Notation

We use i to denote a participant randomized in study j to receive treatment t_{ij} , which can be either 1 or 0, indicating the experimental and control treatment respectively. For each participant,

we have information on a total of p baseline covariates, $\mathbf{x}_{ij}$, where bold indicates a vector. We use n_j to denote the total number of participants randomized in study j . For this participant, we observe a continuous outcome of interest, y_{ij} . We assume that for each participant two underlying outcomes exist, one under treatment and one under control, only one of which is observed. The difference between these two outcomes is the true ITE for the participant i ; this is the quantity of primary interest in this paper. We denote this as B_{ij} (where B stands for benefit). **Table 1** provides an overview of the notations used in this paper.

Table 1: A list of all notations used.

Notation	Description
i	Participant indicator
j	Study indicator
n_j	Number of participants included in study j
p	Number of baseline participant-level covariates
$\mathbf{x}_{ij} = (x_{ij1}, x_{ij2}, \dots, x_{ijp})$	Vector of the p baseline covariates observed for the participant i in study j
t_{ij}	Treatment indicator: 1 for experimental treatment, 0 for control
y_{ij}	Continuous outcome measured on participant i in study j
B_{ij}	True individualized treatment effect (benefit from the treatment) for participant i in study j . This is unobservable.
$\mathbf{x}_{new} = (x_1, x_2, \dots, x_p)$	A vector of p individual-level covariates observed for a new individual (not included in the dataset) for whom we want to predict the treatment effect
M_j	Model fitted in the first stage using IPD from study j
$B_{M_j}^*(\mathbf{x}_{new})$	Predicted individualized treatment effect (benefit from the treatment) for a new individual with covariates $\mathbf{x}_{new}$

2.2 Overview of the framework

We hereby describe a general flexible framework for two-stage IPD meta-analyses with the focus on predicting the ITE for new individuals, so as to inform treatment decisions. At the first stage, we use participant data from the available RCTs and fit a model within each study, using the baseline participant-level covariates and the treatment indicator as predictors. The first stage model can be of any type, ranging from linear regression all the way up to complex machine learning models. For simplicity, in what follows we initially assume that the same model is fitted in each study. In Section 2.4.2 and 2.4.3, we relax this assumption and describe how different models can be used in different studies; we also come back to this point in the Discussion section.

The models fitted at the first stage may either provide predictions about the outcome in both the experimental treatment and the control, with their difference yielding the predicted ITE, or directly predict ITE, without estimating the outcome in the two groups separately; an example of such a model is the causal forest [16–18].

At the second stage, we pool results from the first stage to obtain a final prediction of the individualized treatment effect for a new individual, given their covariates. The pooling method used depends on the type of model fitted at the first stage. In the following sections, we describe in more detail the methods that can be leveraged, and how the various first-stage models can be combined with various second-stage pooling methods.

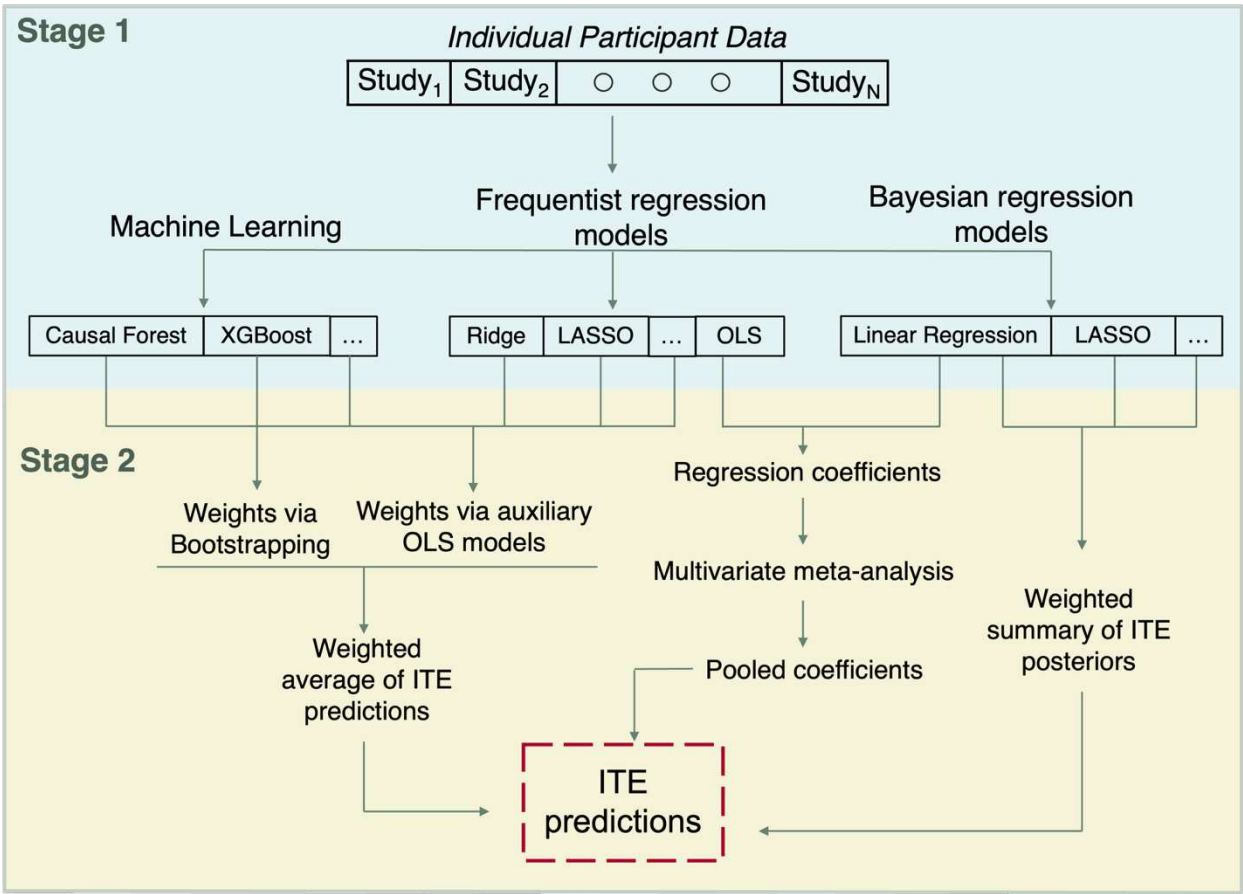
Table 2 provides an overview of all methods described in this paper, while **Figure 1** illustrates how first stage models and second stage are combined within the proposed framework.

Table 2: An overview of the explored two-stage IPD MA methods that can be used to predict ITE.

FREQUENTIST METHODS	
<i>First stage</i>	<i>Second stage</i>
OLS regression	Inverse-variance multivariate meta-analysis of regression coefficients
LASSO	Weighted average of predicted ITE using bootstrap to obtain weights
	Weighted average of predicted ITE using auxiliary OLS regression to obtain weights
Ridge	Weighted average of predicted ITE using bootstrap to obtain weights
	Weighted average of predicted ITE using auxiliary OLS regression to obtain weights
Causal forest (CF)	Weighted average of predicted ITE using bootstrap to obtain weights
	Weighted average of predicted ITE using an auxiliary OLS regression to obtain weights
Extreme Gradient Boosting (XGBoost)	Weighted average of predicted ITE using bootstrap to obtain weights
	Weighted average of predicted ITE using an auxiliary OLS regression to obtain weights
BAYESIAN METHODS	
Bayesian linear model	Bayesian multivariate meta-analysis of regression coefficients
	Weighted summary of ITE posterior distributions
Bayesian LASSO	Weighted summary of ITE posterior distributions

In the first column we provide a list of the various models that can be used at the first stage, where each study is analysed separately. In the second column, we match each first stage model with a second-stage pooling method. **Abbreviations:** OLS: Ordinary Least Squares, ITE: individualized treatment effect.

Figure 1: Visual representation of the framework based on a two-stage IPD-MA to predict individualized treatment effect.



In the upper part of the figure, we outline the first stage, where participant-level data are collected from each study and study-specific models are implemented; these models can include, for example, machine learning approaches, frequentist regression models and Bayesian models. The lower part of the figure displays the pooling strategies that can be used depending on the first stage models. These strategies include the weighted average of ITE predictions, the multivariate meta-analysis of regression coefficients, and the weighted mixing of the posteriors. **Abbreviations:** OLS (Ordinary Least Squares), ITE (Individualized Treatment Effect).

2.3 First stage: analysing IPD data from each study

We describe a range of statistical and machine learning models that can be used for the analysis at the first stage. This list of models is by no means exhaustive; our framework can accommodate any type of statistical or machine learning model used at the first stage, as long as it predicts the ITE.

2.3.1 Linear model: Ordinary least squares (OLS) and Bayesian regression

Linear regression is considered the reference statistical model [19] for continuous outcomes. The outcome is modelled using a simple linear combination of covariates, including treatment covariate-interactions:

$$y_{ij} \sim N(\mu_{ij}, \sigma_j^2)$$
$$\mu_{ij} = \alpha_j + \boldsymbol{\beta}_j \mathbf{x}_{ij} + \boldsymbol{\gamma}_j \mathbf{x}_{ij} t_{ij} + \delta_j t_{ij} \quad (1)$$

In this equation, α_j represents the intercept. Vectors $\boldsymbol{\beta}_j$ and $\boldsymbol{\gamma}_j$ are sets of regression coefficients for the main effects of the covariates on the outcome and the treatment-covariate interactions, respectively. δ_j is the treatment effect when $\mathbf{x}_{ij} = \mathbf{0}$; if the covariates are centered to their mean values, δ_j will correspond to the average treatment effect for a participant with average covariate values. σ_j^2 corresponds to the variance of the residual error.

In practice, we may only include a subset of the covariates $\mathbf{x}_{ij}$ in the interaction terms, i.e. the ones multiplied by $\boldsymbol{\gamma}_j$. This would lead to the distinction between “prognostic factors” (patient covariates that affect the outcome but do not interact with treatment) and “effect modifiers” (covariates that modify the relative treatment effect). For simplicity, we assume here that all the covariates are included as both prognostic factors and effect modifiers.

When fitting this model in a frequentist setting, parameters are estimated by minimizing the residual sum of squares. Since our aim is on treatment effects, for the purposes of this paper we

are interested in δ_j and γ_j ; these two parameters are the only ones needed for predicting the ITE for a new individual.

We can also fit this model in a Bayesian setting. As in all Bayesian analyses, prior distributions must be assigned to all model parameters $(\alpha_j, \beta_j, \gamma_j, \delta_j)$. These can be “vague” or may be based on external information (“informative” priors). As our aim is to present a generic framework, we do not discuss here the specification of priors in detail. Moreover, in what follows we assume that Markov Chain Monte Carlo (MCMC) sampling is used to fit the model. At the end of this stage, we derive study-specific estimates for δ and γ (the parameters required to predict ITE), together with their corresponding variance-covariance matrices.

2.3.2 Penalized regression models

Penalized regression (shrinkage) models are variations of the OLS model, where a penalty term is added to the loss function. Penalization may reduce model complexity, prevent overfitting, and improve predictive performance [20]. Below, we describe three penalized models. Without loss of generality, we penalize only the interaction terms, i.e. γ . One important aspect of penalized models is the intentional introduction of bias into the estimation of regression parameters. As a result, standard error around the estimated parameters is not available, meaning that the usual inverse-variance approach cannot be employed at the second stage. We come back to this point in Section 2.4.

2.3.2.1 Least Absolute Shrinkage and Selection Operator (LASSO)

LASSO estimates [21] the regression parameters (1) by minimizing the following function:

$$\left\{ \sum_{i=1}^{n_j} (y_{ij} - \alpha_j - \beta_j x_{ij} - \gamma_j t_{ij} x_{ij} - \delta t_{ij})^2 + \lambda_j \sum_{l=1}^p |\gamma_{l,j}| \right\}$$

Where λ is a non-negative tuning parameter that controls the amount of shrinkage applied to the parameters related to effect modification. When all $\lambda_j = 0$, the LASSO estimates coincide with OLS. As λ_j increases, the penalized parameters are increasingly shrunk towards zero. When λ_j becomes sufficiently large, some or even all interactions are shrunk to exactly zero. This allows LASSO to perform variable selection, meaning that some of the effect modification parameters are not included in the final model. The optimal value for λ_j is usually decided through k -fold cross-validation.

2.3.2.2 Bayesian LASSO

The Bayesian LASSO is the Bayesian counterpart of LASSO, and utilizes a conditional Laplace prior controlled by the shrinkage parameter λ_j [22] on the interaction parameter:

$$\pi(\gamma_j | \sigma_j^2) = \prod_{l=1}^p \frac{\lambda_j}{2\sqrt{\sigma_j^2}} e^{-\frac{\lambda_j |\gamma_{jl}|}{\sqrt{\sigma_j^2}}}$$

where p is the number of covariates, and λ_j is the parameter that controls the amount of shrinkage. For λ_j we can assign a diffuse hyperprior on λ_j^2 [22,23]. As in the frequentist LASSO, we chose to penalize only the interaction, but we could easily apply the penalization on the prognostic effect and/or the main treatment effects as well, using the same or a different λ_j . For simplicity, we do not explore these extensions further here.

2.3.2.3 Ridge Regression

Ridge regression is another widely used penalization method [24,25]. Ridge estimates the regression parameters by minimizing the following objective function:

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$$\left\{ \sum_{i=1}^n (y_{ij} - \alpha_j - \beta_j x_{ij} - \gamma_j t_{ij} x_{ij} - \delta t_{ij})^2 + \lambda \sum_{l=1}^p \gamma_{l,j}^2 \right\}$$

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In contrast to LASSO, Ridge does not shrink parameters exactly to zero for finite values of λ , and

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therefore does not perform variable selection.

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2.3.3 Extreme Gradient Boosting

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Extreme Gradient Boosting (XGBoost) [26] is a tree-based boosting system, and can be used for

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both binary and continuous outcomes. It is popular due to its ability to handle large and sparse

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datasets. As in most machine learning models, XGBoost incorporates mechanisms to control

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overfitting. First, a shrinkage parameter scales the contribution of each tree, preventing any single

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tree from having an excessive impact on the ensemble. Second, feature subsampling reduces

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variance by fitting trees on a random subset of predictors, an approach originally popularized by

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random forests [27]. XGBoost separately predicts the outcome under treatment and the outcome

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under the control; the difference between the two is the ITE.

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2.3.4 Causal Forests

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Causal Forest [16–18] is a non-parametric machine learning model. It is a generalisation of the

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random forest model, for the case of predicting heterogeneous treatment effects rather than

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absolute outcomes [27]. It is one of the most popular machine learning methods for predicting ITE

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[28]. In contrast to most other statistical or machine learning models, such as the ones described

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above, which predict the outcomes separately under treatment and control, causal forests predict

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the ITE directly, without predicting absolute outcomes under treatment and control. The model

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first creates an ensemble of causal trees, each of which is used to make predictions of the

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treatment effect for a new individual; the predictions from each tree are then combined, to obtain

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a final prediction of the ITE.

2.4 Second stage: methods for pooling results and making predictions

At the second stage of the analysis, we need to synthesize (pool) the study-specific results obtained from the first stage. Depending on the choice of first-stage model, different synthesis methods can be used. **Figure 1** and **Table 2** show the matching of first-stage models with second-stage pooling methods. These are only examples of combinations, but, since the work presented here is a flexible framework, we can use other first-stage models or second-stage methods.

2.4.1 Inverse-variance multivariate meta-analysis of coefficients

This method follows the traditional approach to IPD-MA. It can be readily used when an unpenalized regression is fitted at the first stage, but not with penalized models. In this case, the uncertainty around the estimated parameters, i.e. their standard error, is not evaluated: this leads to the impossibility to use the usual inverse-variance approach.

In more details, for the case of (frequentist) OLS regression: after fitting Equation (1) separately in each study at the first stage, we estimate $\hat{\delta}_j$ and $\hat{\gamma}_j$, along with the corresponding variance-covariance matrix $\hat{\mathbf{S}}_j$. At the second stage, we conduct a multivariate meta-analysis of these $p + 1$ parameters, either with a fixed (common) or random effects model. The general specification of a random effects multivariate meta-analysis of a vector of $p + 1$ parameters, can be written as:

$$\begin{pmatrix} \hat{\gamma}_j \\ \hat{\delta}_j \end{pmatrix} \sim N \left(\begin{pmatrix} \gamma_j \\ d_j \end{pmatrix}, \hat{\mathbf{S}}_j \right)$$

$$\begin{pmatrix} \gamma_j \\ d_j \end{pmatrix} \sim N \left(\begin{pmatrix} \gamma \\ \delta \end{pmatrix}, \mathbf{\Sigma} \right)$$

Where $\mathbf{\Sigma}$ is the between-study variance-covariance matrix of the parameters, representing the between-study heterogeneity. After performing the meta-analysis, we use the pooled estimates of

the parameters $\hat{\gamma}$ and $\hat{\delta}$ to predict the treatment effect for a new individual with covariates $\mathbf{x}_{new}$ as $B^*(\mathbf{x}_{new}) = \hat{\gamma} \mathbf{x}_{new} + \hat{\delta}$.

Similarly, after using an unpenalized Bayesian regression model at the first stage, we can conduct a multivariate meta-analysis within a Bayesian framework at the second stage. For this, we make the assumption of normality of the treatment effect parameter δ and effect modification γ , and assign appropriate prior distributions to the parameters.

2.4.2 Weighted average of study-specific predictions using inverse-variance from auxiliary OLS models

The approach described in the previous paragraph is not applicable to non-parametric first-stage models (such as ML), or models that do not directly provide standard errors (such as penalized regression). In these cases, we still need to synthesize results from the first stage while accounting for the uncertainty of the information provided by each study. As a simple solution, when aiming to predict the ITE for a new individual, we can use a weighted average of the study-specific ITE predictions, where the weights can be calculated as the inverse of the variance of the ITE predictions obtained from auxiliary OLS regression models. More specifically, in study j we fit model M_j (e.g., a penalized regression or a machine learning model); we then use this model to predict the effects for a new subject with covariates $\mathbf{x}_{new}$; let us denote this as $B_{M_j}^*(\mathbf{x}_{new})$. In each study we also fit a separate auxiliary OLS regression model. Using the estimated variance-covariance matrix $\hat{\mathbf{S}}_j$ of $\hat{\gamma}_j$ and $\hat{\delta}_j$ from Equation (1), we calculate the variance of the predicted treatment effect for the new individual according to OLS. This variance is $\hat{u}_j^{OLS}(\mathbf{x}_{new}) = (\mathbf{x}_{new}, \mathbf{1}) \hat{\mathbf{S}}_j (\mathbf{x}_{new}, \mathbf{1})^T$. Finally, we average the $B_{M_j}^*(\mathbf{x}_{new})$ for all studies ($j = 1, \dots, J$), weighted according to the inverse of the corresponding $\hat{u}_j^{OLS}(\mathbf{x}_{new})$. Formally, given the baseline covariates of a new individual $\mathbf{x}_{new}$, the pooled ITE is estimated as:

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$$B_M^*(\mathbf{x}_{new}) = \frac{\sum_{j=1}^J \frac{1}{\hat{u}_j^{OLS}(\mathbf{x}_{new})} B_{M_j}^*(\mathbf{x}_{new})}{\sum_{j=1}^J \frac{1}{\hat{u}_j^{OLS}(\mathbf{x}_{new})}}$$

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In simple words, we meta-analyse the study-specific ITE predictions for each new individual separately, where the weights are the inverse of the variance obtained from an auxiliary OLS regression model. In practice, since data from different studies may be stored in different servers, each study provides the fitted model and the variance-covariance matrix from the corresponding auxiliary OLS model. These results are stored centrally and used to generate pooled predictions on every new individual, given their covariates.

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Since the weights do not depend on the type of the first-stage models, this pooling method is applicable even when different models are used at the first stage. This means, for example, different type of models can be used in different studies, or even different sets of covariates.

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2.4.3 Weighted average of study-specific predictions using bootstrapping

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The method described in the previous paragraph uses the auxiliary OLS variance as a proxy for the uncertainty of the predictions obtained from M_j . An alternative approach is to leverage bootstrapping to estimate this variance directly, thereby avoiding the auxiliary OLS models. In more detail, in study j we fit model M_j , to store centrally. We then generate a large number N_B (e.g., at least $N_B = 100$) of bootstrap samples of the IPD, each of which is used as training data to fit model again; we also extract and store these N_B models centrally. For a new patient with covariates $\mathbf{x}_{new}$, we use model M_j to predict the treatment effect $B_{M_j}^*(\mathbf{x}_{new})$. From the N_B models, we also obtain N_B predictions of this treatment effect; these are solely used to estimate the variability of the prediction $B_{M_j}^*(\mathbf{x}_{new})$. Finally, we calculate a weighted average of the predicted effects for the new subject from the different studies, where the weight is calculated using the N_B bootstrapped versions of M_j . Let $B_{M_{j,s}}^*(\mathbf{x}_{new})$ denote the prediction obtained from the model $M_{j,s}$.

245 fitted on the bootstrap sample s ($s = 1, \dots, N_B$) in study j , and $\bar{B}_{M_j,BS}(\mathbf{x}_{new})$ the mean of these
 246 bootstrap predictions. We then calculate the weights of each study j , using the N_B bootstrapped
 247 versions of M_j :

$$248 \quad \hat{u}_j^{BS}(\mathbf{x}_{new}) = \frac{1}{N_B} \sum_{s=1}^{N_B} \left(B_{M_j,s}^*(\mathbf{x}_{new}) - \bar{B}_{M_j,BS}(\mathbf{x}_{new}) \right)^2$$

249 Finally, we calculate a weighted average of the predicted effects for the new individual:

$$250 \quad B^{*M}(\mathbf{x}_{new}) = \frac{\sum_{j=1}^J \frac{1}{\hat{u}_j^{BS}(\mathbf{x}_{new})} B^{*M_j}(\mathbf{x}_{new})}{\sum_{j=1}^J \frac{1}{\hat{u}_j^{BS}(\mathbf{x}_{new})}}$$

251 Note that this approach is very generic and, as for the OLS-based inverse variance, it can be
 252 used even when different first-stage models are fitted across studies.

253 2.4.4 Weighted mixing of Bayesian ITE posterior samples

254 From the Bayesian models fitted at the first stage in each study separately, we extract the
 255 posterior samples of δ and γ and store them centrally, i.e., in a common server. Then, when we
 256 want to predict the ITE for a new individual, we apply these samples to obtain a posterior of ITE
 257 from each study separately. Subsequently, we follow a weighted mixing approach to synthesize
 258 across studies.

259 In more detail, suppose that after the first stage analysis, from study j we retain a large number
 260 of MCMC samples, for example 10,000 draws, for δ_j and γ_j . Storing the posteriors of δ_j and γ_j
 261 centrally is required, to be able to repeat the analyses described below for every new individual
 262 that we want to make predictions for. For each MCMC sample, and given the new individual's
 263 baseline covariates $\mathbf{x}_{new}$, we compute the ITE as $\delta_j + \gamma_j \mathbf{x}_{new}$ from study j , and obtain the
 264 corresponding posterior distribution. From these 10,000 samples we calculate the mean,
 265 $\overline{B_j^*}(\mathbf{x}_{new})$, and the corresponding sampling variance, $Var(B_j^*(\mathbf{x}_{new}))$. We repeat this procedure for

all studies. At the end of this process, from each study we have a predicted ITE for the new individual, and a measure of uncertainty around it. Finally, we calculate a weighted average of the study specific predictions, using weights equal to the inverse of their posterior variance.

In essence this procedure resembles the inverse-variance approach: however, instead of synthesizing the estimates of the regression parameters, we directly synthesize predicted the individual-level ITEs.

2.5 Description of the simulation study

Here we present the simulation study following the guidance by Morris et al. [29].

2.5.1 Aim

We aimed to compare, within a simulation study, all combinations of first-stage models and second-stage pooling methods described in Section 2 and outlined in **Table 2**, for the purpose of predicting the ITE of new individuals.

2.5.2 Data generation mechanism

We simulated IPD from RCTs under 16 different data generating mechanism (DGM) scenarios. For each scenario, we generated 100 simulated datasets. In each dataset, we simulated IPD from 10 studies. The parameters that varied across scenarios were as follows:

- The sample size of the RCTs, drawn either extracted from a uniform distribution $U(50, 200)$ ('small sample size') or $U(300, 600)$ ('large sample size').
- The magnitude of the effect modification, i.e., the treatment-covariate interaction (γ , small vs large).
- The extend of heterogeneity of treatment effect across studies (τ_{δ}^2 , small vs large)
- The type of treatment-covariate interactions (linear vs 'complex', i.e., non-linear)

To simulate the data, we followed the steps below. For each study j , we simulated:

- The sample size n_j .
- An intercept per study, $\alpha_j \sim N(2, 0.5)$, corresponding to the average outcome in participants treated with the control and all covariates set to 0.
- The treatment assignment t_{ij} for participant i in study j , drawn from a Bernoulli distribution with probability 0.5 ($p = 0.5$).
- The study-specific treatment effect for a participant with all covariates set to 0; this was drawn from $\delta_j \sim N(2, \tau^2)$, where τ^2 is heterogeneity, defined by the scenario.
- Eight participant-level baseline covariates $\mathbf{x}_{ij}$, four of which are continuous and four are dichotomous;
- A participant-level residual error, drawn from $\epsilon_{ij} \sim N(0, \sigma^2)$, where $\sigma = 0.5$.

For scenarios with linear DGMs for the treatment-covariate interactions, the outcome for participant i in study j was generated via the model $y_{ij} = \alpha_j + \boldsymbol{\beta} \mathbf{x}_{ij} + \boldsymbol{\gamma} \mathbf{x}_{ij} t_{ij} + \delta_j t_{ij} + \epsilon_{ij}$, where $\boldsymbol{\beta}$ is the vector of coefficients for the main effects of the covariates, $\boldsymbol{\gamma}$ the vector of coefficients for effect modification (treatment-covariate interactions) and δ_j the average treatment effect. In scenarios with complex interactions, an extra non-linear interaction term was added to the DGM. Details are given in **Appendix 1**.

For each scenario, we simulated IPD from 10 studies; this was the training dataset, i.e., the data we used to develop all models. We also used the same DGM to generate data from another 3 studies, with 1'000 individuals each; this was the testing dataset, which was used to assess all models' performance (as described below). **Table 3** provides an overview of all scenarios we explored.

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Table 3: An overview of the scenarios we explored

Scenario	Interaction	Sample size	Effect modification	Heterogeneity
1	Linear	Small	Small	Small
2	Linear	Small	Small	Large
3	Linear	Small	Large	Small
4	Linear	Small	Large	Large
5	Linear	Large	Small	Small
6	Linear	Large	Small	Large
7	Linear	Large	Large	Small
8	Linear	Large	Large	Large
9	Complex	Small	Small	Small
10	Complex	Small	Small	Large
11	Complex	Small	Large	Small
12	Complex	Small	Large	Large
13	Complex	Large	Small	Small
14	Complex	Large	Small	Large
15	Complex	Large	Large	Small
16	Complex	Large	Large	Large

313

314 2.5.3 Target quantity

315 The quantity of interest was the true ITE for each individual in the testing dataset. This was exactly
 316 known for each individual in the testing dataset, given the DGM we used for each scenario and
 317 the individual's covariates.

318 2.5.4 Models compared / implementation details

319 We used the training dataset (10 simulated RCTs per simulation) to develop all models described
 320 in Section 2 and **Table 2**. For models requiring tuning of parameters and/or hyperparameters, i.e.,
 321 the shrinkage regression models and XGBoost, optimal values were obtained through 10-fold
 322 internal cross validation. Further details about the models' implementations, including Bayesian
 323 analyses and parameters tuning, are provided in **Appendix 2-5**.

324 2.5.5 Measures of performance

325 We used the models that we developed in the training dataset (see previous paragraph) to make
 326 predictions about all individuals included in the testing dataset. We then compared the true ITE
 327 of each individual from the simulated data (B , known via the DGM) with the predictions obtained

from the models (B^*) and assessed the performance of each model using the Median Absolute Error, i.e., $MAE = \text{median}(|B - B^*|)$, the mean bias, i.e., $\text{bias} = \text{mean}(B - B^*)$, and the coefficient of determination R^2 of the regression of the true ITEs on the predictions, i.e., the OLS $B \sim b_0 + b_1 B^*$.

2.5.6 Software

All analyses were performed in R (v. 4.5.0) [30]. We used the packages **dplyr** [31], **glue** [32], **mvtnorm** [33] **matrixStats** [34] for data generation and manipulation. The following packages were used for model fitting: **mvmeta** [35] (multivariate meta-analysis), **glmnet** [36] (LASSO and Ridge), **grf** [37] (causal forest), **caret** [38] and **xgboost** [39] (XGBoost), and **rjags** [40] (Bayesian analysis). Finally, parallel computation was implemented using **doParallel** [41] and **parallel** [30]. Further details on the analyses, all data, and all R codes used are provided in the online appendix and our GitHub repository <https://github.com/esm-ism-unibe-ch-REPRODUCIBLE/IPD-MA-IndividualizedTreatmentEffect>.

3 Simulation study results

Table 4 summarizes the performance measures of the different combinations of first-stage study-specific models and second-stage pooling methods across the simulated scenarios. Results are also shown, in **Figure 2** (MAE), **Figure 3** (mean bias) and **Figure 4** (R^2). Below we summarize the main findings from the simulations.

3.1 OLS vs shrinkage-based approaches

Shrinkage methods, such as LASSO and Ridge regression, generally showed modest improvements over the OLS-based inverse-variance meta-analysis (OLS+MVMA), particularly in scenarios with small sample sizes. For example, in scenario 3, the median MAE was 1.86 for

OLS+MVMA and 1.83 for LASSO combined with inverse bootstrap variance (LASSO+BS). However, these improvements were not consistently observed across all scenarios, particularly in the presence of large between-study heterogeneity.

3.2 Statistical vs machine learning models

In scenarios with linear effect modifications, regression-based models generally performed better than machine learning approaches, especially when sample sizes were small. ML models showed generally limited predictive performance in scenarios with small sample sizes, both under linear and complex interactions (scenarios 1–4 and 9–12). For instance, in Scenario 2, the median MAE was 0.52 for OLS+MVMA, compared with values larger than 1.24 for all ML models explored. As sample sizes increased, the difference in performance was smaller: in scenario 6, the median MAE for OLS+MVMA was 0.46, compared with 0.87 for the causal forest with OLS-based inverse variance (CF+AuxOLS) and 0.75 for XGBoost with OLS-based inverse variance (XGB+AuxOLS).

Conversely, in scenarios with complex interactions and large sample sizes (scenarios 13-16), ML models clearly outperformed regression-based models. For example, in Scenario 13 the median MAE was 1.07 with XGB+AuxOLS and 1.86 for OLS-MVMA; R^2 was similarly better for XGBoost (0.75 for XGB+AuxOLS vs 0.49 for OLS-MVMA). At the same time, the bias was larger than the one for OLS+MVMA, regardless of the pooling method, resulting in median values across simulations for this scenario of -0.30 and -0.06, for the two methods respectively. This finding reflects the different bias-variance trade-off of the two models. This is illustrated in **Figure S1** of **Appendix 6** for one dataset generated under scenario 13; OLS+MVMA showed a large variance (MAE) in predictions, while maintaining a low bias, whereas XGB+AuxOLS displayed a lower variance of predictions but with larger bias. Bias patterns varied across methods and scenarios, with some machine learning approaches showing substantial negative bias in complex settings of the treatment-covariates interactions.

3.3 Performance of Bayesian approaches

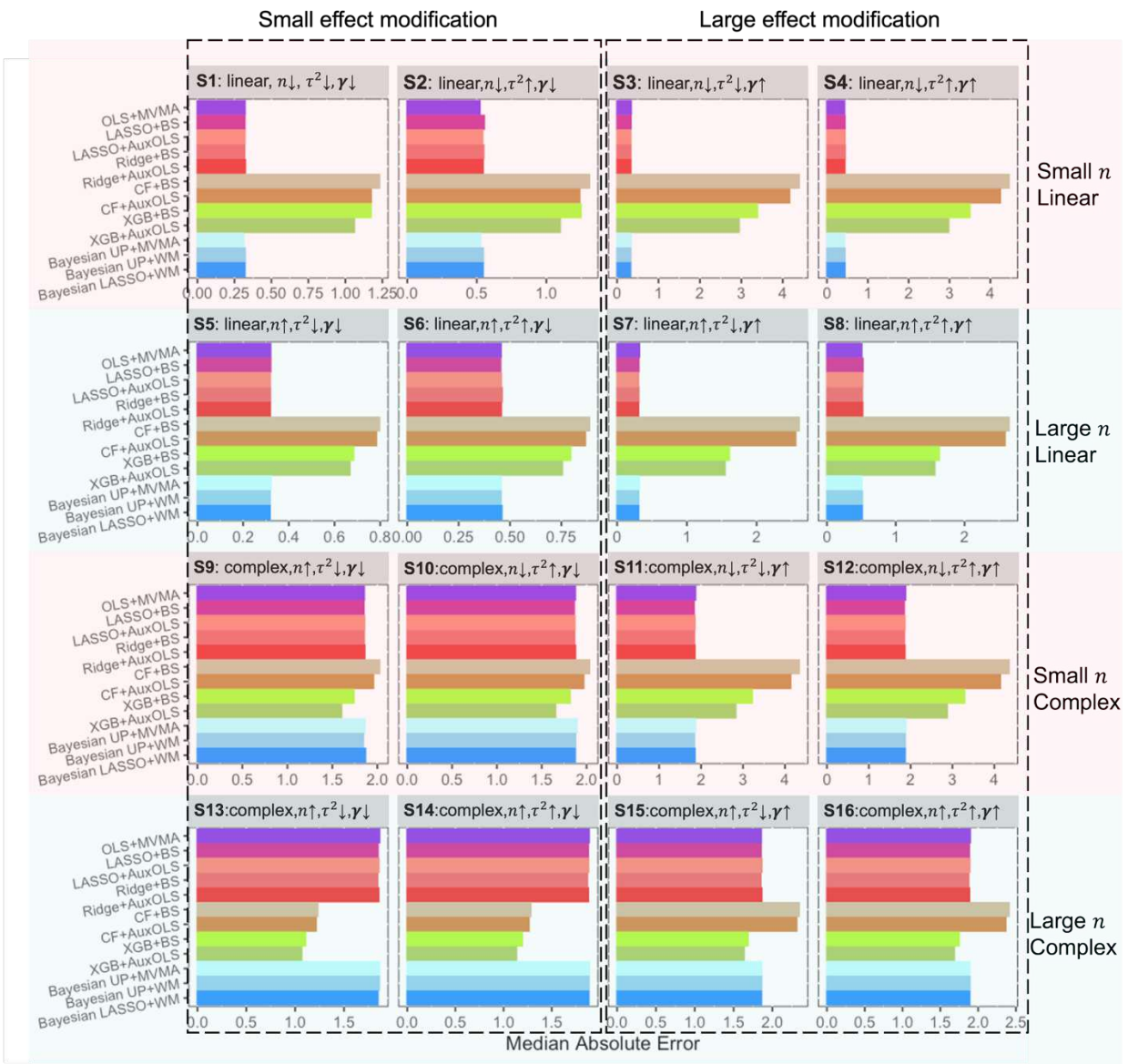
Frequentist (OLS+MVMA) and Bayesian multivariate meta-analyses (Bayesian UP+MVMA) performed similarly across scenarios in terms of MAE and R^2 . Comparing the three Bayesian approaches, Bayesian UP+MVMA appeared to predict ITEs slightly more accurately than the weighted mixing of the MCMC posterior samples, in scenarios with small sample sizes and large heterogeneity, for both the Bayesian linear regression (Bayesian ITE-LM) and Bayesian LASSO (Bayesian ITE-LASSO). For instance, in scenario 2, the median MAE of the Bayesian UP+MVMA was 0.52 versus 0.55 for Bayesian ITE-LM and 0.55 for Bayesian ITE-LASSO. In contrast, under complex interactions, the three Bayesian methods performed similarly. Further, Bayesian ITE-LM and Bayesian ITE-LASSO showed similar performances across scenarios.

3.4 Comparison of weighting methods

Overall, weighting based on the inverse variance from auxiliary OLS models appeared to perform better in terms of R^2 and MAE, across first stage models. For shrinkage models, differences between the two pooling approaches were minimal, whereas for causal forest and XGBoost, auxiliary OLS-based weighting tended to outperform bootstrap-based weighting. For example, in scenario 6, XGBoost combined with OLS-based inverse variance (XGB+AuxOLS) model gave a MAE of 0.75, while bootstrapping (XGB+BS) gave 0.80.

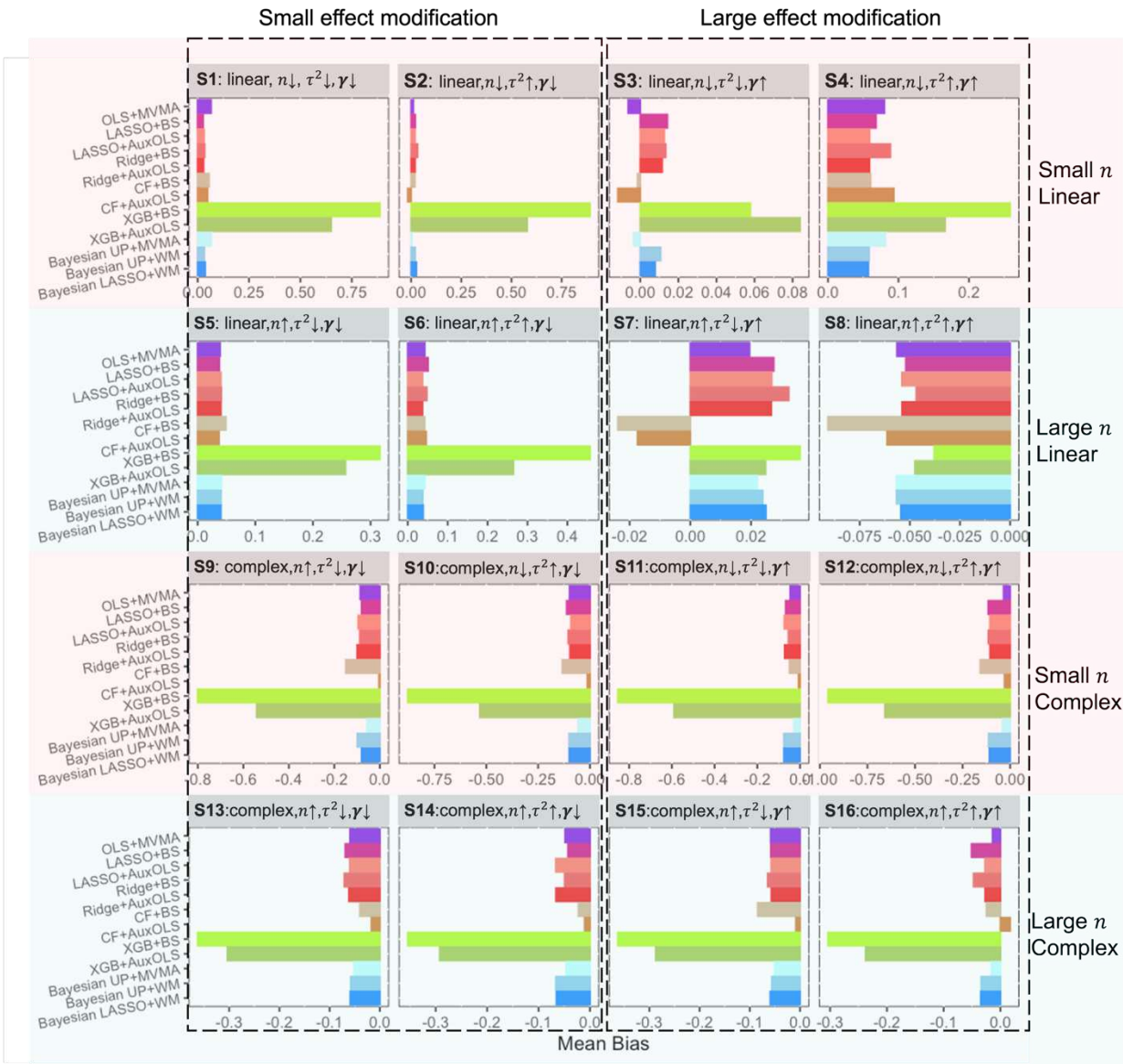
[INSERT TABLE 4 HERE]

Figure 2: Barplots of the median MAE across 100 iterations of simulated data for each method.



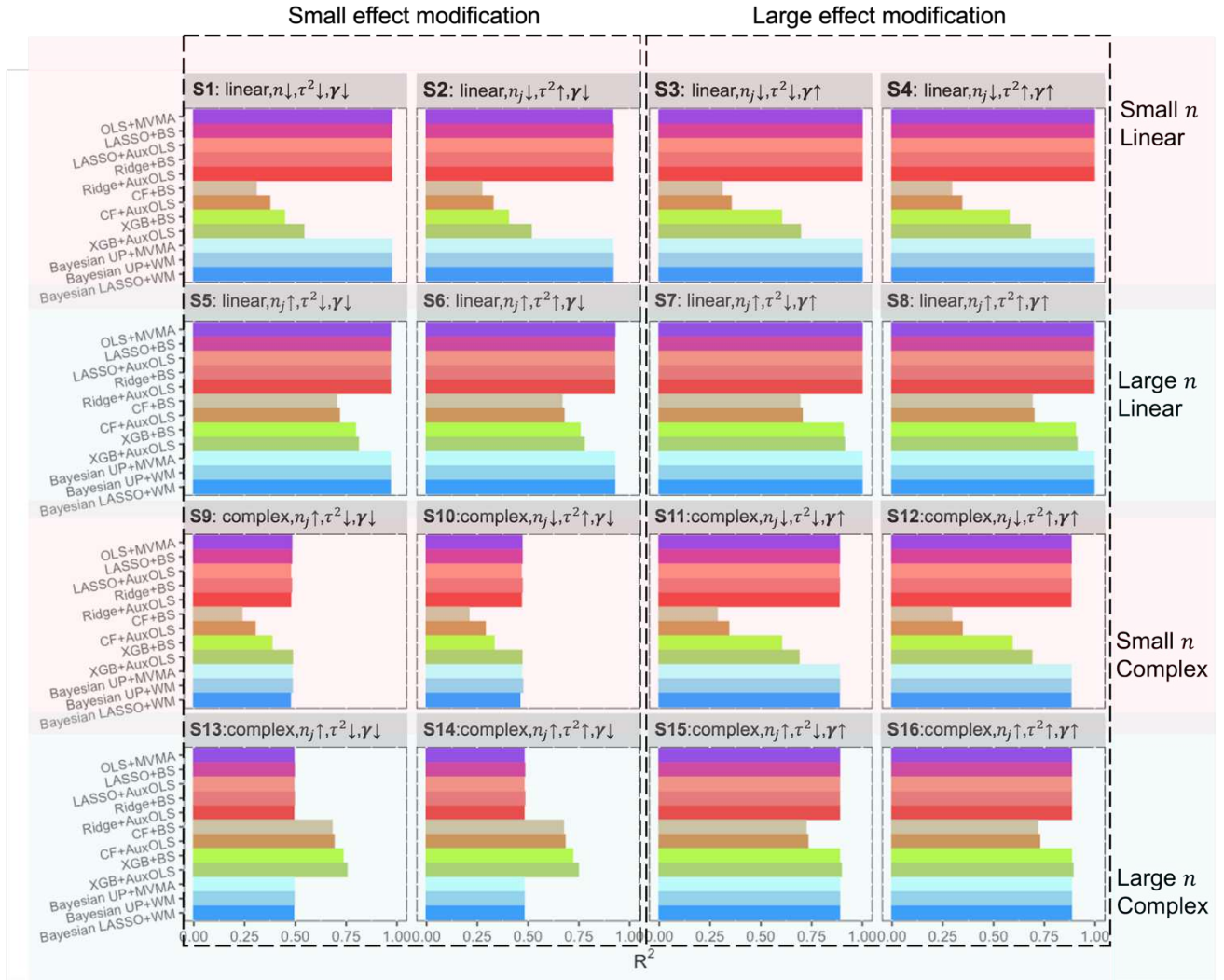
Each panel corresponds to a different scenario (1-16). The combined approaches are noted as (1st stage model)+(2nd stage pooling method), following abbreviations as per Table 4. Abbreviations: MAE (Median Absolute Error)

Figure 3: Barplots of the median mean bias across 100 iterations of simulated data for each method.



Each panel (a-r) corresponds to a different scenario (1-16). The combined approaches are noted as (1st stage model)+(2nd stage pooling method), following abbreviations as per Table 4.

Figure 4: Barplots of the determination coefficient (R^2) across 100 iterations of simulated data for each method.



Each panel (a-r) corresponds to a different scenario (1-16). The combined approaches are noted as (1st stage model)+(2nd stage pooling method), following abbreviations as per Table 4.

4 Discussion

In this work, we focused on methods for performing a two-stage individual participant data meta-analysis (IPD-MA), when the aim is the prediction of individualized treatment effects (ITEs) in new individuals. A two-stage approach is required, for example, when the data from different RCTs are

stored on separate servers and cannot be combined into a single dataset for joint analyses. Our objective was to present a generalized framework for performing such analyses going beyond the currently available methods, which are typically based on simple regression models (at the first stage) combined with an inverse variance approach (at the second stage).

More specifically, we described a range of different statistical and machine learning models that can be used at the first stage, including linear regression (OLS) as a reference model, penalized regression models such as LASSO and Ridge, and also machine learning models, such as the causal forest and XGBoost. In addition, we also considered Bayesian regression models, i.e., linear regression model with vague priors and Bayesian LASSO. We then presented alternative second-stage approaches to pool the study-specific outputs from the first stage, in order to predict ITEs on new individuals. These included: (a) the frequentist inverse-variance multivariate meta-analysis of regression coefficients; (b) the weighted average of study-specific ITE predictions, with weights obtained via a bootstrap procedure or the using auxiliary OLS regression models; for Bayesian models, we discussed (c) Bayesian multivariate meta-analysis of regression coefficients, and (d) a weighted mixing of ITE posterior distributions, with weights based on inverse variance.

We performed an extensive simulation to evaluate several combinations of first-stage models and second-stage pooling methods across a range of simulated scenarios. As expected, our findings indicated that in scenarios with linear treatment-covariate interactions, regression-based models consistently outperformed machine learning approaches. Conversely, in scenarios with complex effect modifications and larger samples, machine learning approaches performed better in terms of absolute error, although they suffered from larger bias; this is related to the bias-variance trade off [42]. We also found that shrinkage methods provided modest improvements over OLS, mainly in small-sample scenarios; however, no difference was seen in scenarios with large between-study heterogeneity. These results somewhat differ from the findings of Seo et al. [44], who,

instead, recommended shrinkage models in a one-stage IPD-MA setting. Frequentist meta-analyses in general performed similarly to their Bayesian counterpart. This agrees with the findings by Tanriver-Ayder et al. [45]. Across all scenarios, inverse-variance weighting based on OLS appeared to be the best performing pooling strategy.

A key strength of the proposed framework is its flexibility. The methods described can accommodate a wide range of first stage models, including models not assessed in our simulations. For example, support vector machines [46], further gradient boosting methods [47], neural networks [48–50] for more complex data, and additional Bayesian models such as Stochastic Search Variable Selections (SSVS) [51] or Bayesian additive regression tree (BART) [52], could be easily implemented in our setup, as first stage models. The framework can also be extended to settings, in which different first-stage are used in different studies. The framework described here could be easily adapted so as to use at the first stage more advanced models, e.g. causal forest, in larger studies, and regression-based models in smaller studies while still pooling ITE study-specific predictions at the second stage, for example via bootstrapping or using auxiliary OLS models. A further strength of the framework described here is that the second-stage pooling methods can overcome differences in the different sets of variables selected by first-stage models. For example, if variable selection is performed in the first stage, using models such as LASSO, we may have different sets of predictors being selected in the different RCTs; this precludes the use of methods that directly synthesize regression coefficients. By pooling predictions instead of model coefficients, the methods described here remain applicable regardless of the specific predictors retained in each study.

One limitation of our study is that we only explored a small number of scenarios and considered only a limited number of first stage models. Broader simulations could identify additional patterns of model performance, while alternative models not explored here (e.g., elastic net regression [54], other machine learning approaches, or Bayesian approaches such as the SSVS [51]) might

perform better in specific settings. However, our aim was not to identify the overall single best model. Rather, we aimed to show how the proposed framework can accommodate alternative approaches and how the best performing method may depend on the underlying scenario. In practice, when performing a real IPD-MA, the data generating mechanism is unknown, making it impossible to choose one specific method a priori [9,12]. Instead, appropriate internal validation should be used to compare candidate approaches, after evaluating their predictive performance [55–58]. Further, a limitation of some of the methods that we described (e.g., the ones based on bootstrapping) is that they can be computationally demanding to fit at the first stage, particularly for larger datasets.

Future work could extend our framework to cover binary and time-to-event outcomes. The framework could also be generalized to a network meta-analytical setting, in which multiple treatments are compared, leveraging both direct and indirect evidence [59–61].

5 Conclusion

We described a general framework for synthesizing IPD from multiple trials using a two-stage method, when the aim is to predict individual-level treatment effects. Such predictions can facilitate personalized treatment choices and shared medical decision-making in clinical practice.

List of abbreviations

BS:	Bootstrap
CF:	Causal Forest
DGM:	Data Generation Mechanism
IPD-MA:	Individual Participant Data Meta-Analysis
ITE:	Individualized Treatment Effect
IV:	Inverse Variance

497 LASSO: Least Absolute Shrinkage and Selection Operator

498 LM: Linear Model

499 MAE: Median Absolute Error

500 MCMC: Markov Chain Monte Carlo

501 OLS: Ordinary Least Squares

502 RCT: Randomized Controlled Trial

503 XGBoost: Extreme Gradient Boosting

504 Declarations

505 Ethics approval and consent to participate

506 Not applicable

507 Consent for publication

508 Not applicable

509 Availability of data and materials

510 The datasets generated and analyzed during the current study, together with all files necessary
511 to reproduce the analyses, are available in the following GitHub repository
512 <https://github.com/esm-ispm-unibe-ch-REPRODUCIBLE/IPD-MA-IndividualizedTreatmentEffect>.

513 Competing interests

514 None of the authors have any competing interests to disclose.

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518 Authors' contribution

519 Mainetti M: Formal analysis, Methodology, Software, Visualization, Writing – Original Draft.

520 Chalkou K: Methodology, Writing – review & editing. Philipona FL: Writing – review & editing.

521 Schwarzer G: Writing – review & editing. Salanti G: Methodology, Funding Acquisition, Writing –

522 review & editing. Efthimiou O: Conceptualization, Supervision, Funding Acquisition, Methodology,

523 Writing – review & editing.

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Table 4: Median performances: combinations of first-stage models and second-stage pooling-methods across scenarios (MAE, Mean Bias, R^2).

1 st stage model	OLS	LASSO		Ridge		CF		XGBoost		Bayesian UP	Bayesian UP	Bayesian LASSO
2 nd stage pooling method	MVMA	BS	AuxOLS	BS	AuxOLS	BS	AuxOLS	BS	AuxOLS	MVMA	WM	WM
<i>SIMULATED SCENARIO</i>												
1. Linear Interactions	0.32	0.32	0.32	0.32	0.32	1.23	1.17	1.17	1.06	0.31	0.32	0.32
Small sample size	0.07	0.03	0.03	0.03	0.03	0.06	0.05	0.89	0.65	0.07	0.03	0.04
Small effect modification	0.97	0.97	0.97	0.97	0.97	0.31	0.37	0.45	0.54	0.97	0.97	0.97
Small heterogeneity												
2. Linear Interactions	0.52	0.55	0.54	0.55	0.54	1.31	1.24	1.25	1.10	0.52	0.55	0.55
Small sample size	0.01	0.02	0.02	0.03	0.02	0.02	-0.02	0.89	0.58	0.004	0.02	0.03
Small effect modification	0.92	0.92	0.92	0.92	0.92	0.27	0.33	0.40	0.51	0.91	0.92	0.92
Large heterogeneity												
3. Linear Interactions	0.34	0.33	0.33	0.33	0.33	4.39	4.16	3.39	2.94	0.33	0.32	0.32
Small sample size	-0.01	0.01	0.02	0.01	0.01	-0.001	-0.01	0.06	0.08	-0.003	0.01	0.01
Large effect modification	>0.99	>0.99	>0.99	>0.99	>0.99	0.31	0.35	0.60	0.69	>0.99	>0.99	>0.99
Small heterogeneity												
4. Linear Interactions	0.43	0.43	0.44	0.44	0.43	4.45	4.24	3.50	2.98	0.43	0.43	0.43
Small sample size	0.08	0.07	0.06	0.09	0.06	0.06	0.09	0.26	0.17	0.08	0.06	0.06
Large effect modification	>0.99	>0.99	>0.99	0.99	>0.99	0.29	0.34	0.58	0.68	>0.99	>0.99	>0.99
Large heterogeneity												
5. Linear Interactions	0.32	0.32	0.32	0.32	0.32	0.80	0.78	0.68	0.67	0.32	0.32	0.32
Large sample size	0.04	0.04	0.04	0.04	0.04	0.05	0.04	0.32	0.26	0.04	0.04	0.04
Small effect modification	0.97	0.97	0.97	0.97	0.97	0.70	0.72	0.79	0.809	0.97	0.97	0.97
Small heterogeneity												
6. Linear Interactions	0.46	0.45	0.46	0.46	0.46	0.89	0.87	0.80	0.75	0.46	0.46	0.46
Large sample size	0.04	0.05	0.04	0.05	0.04	0.04	0.05	0.46	0.27	0.04	0.04	0.04
Small effect modification	0.93	0.92	0.93	0.93	0.93	0.67	0.68	0.75	0.78	0.93	0.93	0.93
Large heterogeneity												
7. Linear Interactions	0.32	0.31	0.31	0.31	0.31	2.61	2.56	1.61	1.55	0.32	0.31	0.31
Large sample size	0.02	0.03	0.03	0.03	0.03	-0.02	-0.02	0.04	0.03	0.02	0.02	0.02
Large effect modification	>0.99	>0.99	>0.99	>0.99	>0.99	0.69	0.70	0.90	0.91	>0.99	>0.99	>0.99
Small heterogeneity												
8. Linear Interactions	0.50	0.52	0.52	0.51	0.52	2.65	2.59	1.63	1.57	0.51	0.51	0.51
Large sample size	-0.06	-0.05	-0.05	-0.05	-0.05	-0.09	-0.06	-0.04	-0.05	-0.06	-0.06	-0.01
Large effect modification	0.99	0.99	0.99	0.99	0.99	0.69	0.67	0.90	0.91	0.99	0.99	0.99
Large heterogeneity												

9. <i>Complex Interactions</i>	1.85	1.85	1.86	1.85	1.86	2.02	1.96	1.74	1.60	1.86	1.84	1.87
<i>Small sample size</i>	-0.09	-0.08	-0.10	-0.88	-0.10	-0.15	-0.004	-0.81	-0.54	-0.06	-0.10	-0.08
<i>Small effect modification</i>	0.48	0.48	0.48	0.48	0.48	0.24	0.30	0.38	0.48	0.48	0.48	0.47
<i>Small heterogeneity</i>												
10. <i>Complex Interactions</i>	1.87	1.86	1.87	1.87	1.87	2.03	1.97	1.81	1.65	1.89	1.88	1.87
<i>Small sample size</i>	-0.10	-0.11	-0.09	-0.11	-0.10	-0.13	-0.01	-0.88	-0.53	-0.06	-0.10	-0.10
<i>Small effect modification</i>	0.47	0.47	0.47	0.47	0.466	0.21	0.29	0.33	0.47	0.47	0.47	0.46
<i>Large heterogeneity</i>												
11. <i>Complex Interactions</i>	1.86	1.83	1.85	1.84	1.85	4.34	4.13	3.21	2.82	1.86	1.84	1.85
<i>Small sample size</i>	-0.05	-0.07	-0.07	-0.05	-0.07	-0.05	-0.01	-0.85	-0.59	-0.03	-0.08	-0.08
<i>Large effect modification</i>	0.89	0.89	0.88	0.89	0.88	0.28	0.34	0.60	0.69	0.89	0.89	0.89
<i>Small heterogeneity</i>												
12. <i>Complex Interactions</i>	1.87	1.84	1.86	1.85	1.86	4.35	4.14	3.29	2.87	1.88	1.87	1.87
<i>Small sample size</i>	-0.03	-0.12	-0.11	-0.11	-0.11	-0.16	-0.03	-0.96	-0.66	-0.04	-0.09	-0.11
<i>Large effect modification</i>	0.88	0.88	0.88	0.88	0.88	0.29	0.34	0.56	0.69	0.88	0.88	0.88
<i>Large heterogeneity</i>												
13. <i>Complex Interactions</i>	1.86	1.85	1.86	1.84	1.85	1.23	1.21	1.11	1.07	1.86	1.86	1.85
<i>Large sample size</i>	-0.06	-0.07	-0.06	-0.07	-0.06	-0.04	-0.02	-0.36	-0.30	-0.05	-0.06	-0.06
<i>Small effect modification</i>	0.49	0.49	0.49	0.49	0.49	0.68	0.690	0.73	0.75	0.49	0.49	0.49
<i>Small heterogeneity</i>												
14. <i>Complex Interactions</i>	1.89	1.88	1.88	1.87	1.88	1.28	1.26	1.19	1.13	1.89	1.88	1.88
<i>Large sample size</i>	-0.05	-0.04	-0.07	-0.05	-0.07	-0.02	-0.01	-0.35	-0.29	-0.05	-0.07	-0.07
<i>Small effect modification</i>	0.48	0.48	0.48	0.48	0.48	0.67	0.68	0.72	0.75	0.48	0.48	0.48
<i>Large heterogeneity</i>												
15. <i>Complex Interactions</i>	1.86	1.85	1.86	1.85	1.86	2.34	2.31	1.68	1.63	1.86	1.86	1.86
<i>Large sample size</i>	-0.06	-0.06	-0.06	-0.06	-0.06	-0.08	-0.01	-0.36	-0.29	-0.05	-0.06	-0.06
<i>Large effect modification</i>	0.89	0.89	0.89	0.89	0.89	0.72	0.73	0.89	0.89	0.89	0.89	0.89
<i>Small heterogeneity</i>												
16. <i>Complex Interactions</i>	1.89	1.88	1.89	1.87	1.88	2.41	2.36	1.75	1.68	1.89	1.89	1.89
<i>Large sample size</i>	-0.01	-0.05	-0.03	-0.05	-0.03	-0.03	0.02	-0.30	-0.24	-0.02	-0.03	-0.04
<i>Large effect modification</i>	0.88	0.88	0.88	0.88	0.88	0.72	0.73	0.88	0.89	0.88	0.88	0.88
<i>Large heterogeneity</i>												

Abbreviations: MAE (Median Absolute Error). First stage models – OLS (Ordinary Least Squares), LASSO (Least Absolute Shrinkage and Selection Operator), CF (causal forest), XGBoost (Extreme Gradient Boosting), Bayesian UP (Bayesian unpenalized regression). Second stage pooling methods – MVMA (multivariate meta-analysis), BS (Inverse Variance meta-analysis of predicted ITE based on weights obtained from bootstrapping, AuxOLS (Inverse Variance meta-analysis of predicted ITE based on weights obtained from an auxiliary OLS model); WM (weighted mixing of MCMC posterior samples of ITE from Bayesian model).

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